BULGARIAN NATIONAL OLYMPIAD IN MATHEMATICS

Third round

1996

Problem 1. Prove that for all positive integers $n \geq 3$ there exist an odd positive integers x_n and y_n, such that

$$7x_n^2 + y_n^2 = 2^n.$$

Problem 2. The circles k_1 and k_2 with centers O_1 and O_2 respectively are externally tangent at the point C, while the circle k with center O is externally tangent to k_1 and k_2. Let ℓ be the common tangent of k_1 and k_2 at the point C and let AB be the diameter of k, which is perpendicular to ℓ, the points A and O_1 lie in one and the same semiplane with respect to the line ℓ. Prove that the lines AO_2, BO_1 and ℓ have a common point.

Problem 3. a) Find the maximal value of the function $y = |4x^3 - 3x|$ in the interval $[-1, 1]$.

b)Let a, b and c be real numbers and M be the maximal value of the function $y = |4x^3 + ax^2 + bx + c|$ in the interval $[-1, 1]$. Prove that $M \geq 1$. For which a, b, c is the equality reached?

Problem 4. The real numbers $a_1, a_2, \ldots, a_n$ $(n \geq 3)$ form an arithmetic progression. There exists a permutation $a_{i_1}, a_{i_2}, \ldots, a_{i_n}$ of the same numbers, which is a geometric progression. Find the numbers $a_1, a_2, \ldots, a_n$, if they are pairwise different and the biggest among them is equal to 1996.

Problem 5. A convex quadrilateral $ABCD$, for which $\angle ABC + \angle BCD < 180°$, is given. The common point of the lines AB and CD is E. Prove that $\angle ABC = \angle ADC$ if and only if

$$AC^2 = CD \cdot CE - AB \cdot AE.$$

Problem 6. A rectangle $m \times n$ $(m > 1, n > 1)$ is divided into mn squares 1×1 with lines, parallel to its sides. In how many ways could two of the squares be canceled and the remaining part be covered with dominoes 2×1?

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